Project Name

Organic Farming Awareness and Product Platform

Software Requirements Specification

Course Code: INT252

Course Name: WEB APP DEVELOPMENT WITH REACTJS

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Roll No: 51 **Section : K23PA**

Prepared for
Continuous Assessment 2
Spring 2025

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1. Introduction

The Organic Farming Awareness and Product Platform is a web-based system designed to promote sustainable agriculture and provide a digital marketplace for organic products. This Software Requirement Specification (SRS) document defines all functional and non-functional requirements needed to design and develop the system.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to clearly define the functional and non-functional requirements of the Organic Farming Awareness and Product Platform. This document is intended for students, instructors, developers, and evaluators involved in the design, development, testing, and assessment of the project.

1.2 Scope

The Organic Farming Awareness and Product Platform is a web-based application developed using React.js (frontend) and Node.js with Express (backend). The system provides:

- *Awareness and educational content about organic farming*
- *Registration and verification of organic farmers and experts (Proposers)*
- *Online marketplace for buying organic products*
- *Admin validation and management system*

The platform does not handle offline payments or logistics management in its initial version.

1.3 Definitions, Acronyms, and Abbreviations

SRS – Software Requirement Specification
UI – User Interface
DFD – Data Flow Diagram
API – Application Programming Interface

1.4 References

IEEE SRS Guidelines
React.js Documentation
Node.js Documentation
MongoDB Documentation

1.5 Overview

This document describes system features, constraints, interfaces, and requirements. It is organized into introduction, general description, specific requirements, and analysis models.

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2. General Description

This section of the SRS should describe the general factors that affect 'the product and its requirements. It should be made clear that this section does not state specific requirements; it only makes those requirements easier to understand.

2.1 Product Perspective

The system is a standalone web application following a client-server architecture. React.js is used for the client-side interface, while Node.js and Express handle server-side logic with MongoDB as the database.

2.2 Product Functions

- *User Registration & Login*
- *Admin Verification*
- *Product Listing*
- *Organic Education Content*
- *Order Management*

2.3 User Characteristics

Admin – Technical knowledge
Farmers – Basic computer skills
Users – General public

2.4 General Constraints

- *Internet Required*
- *Web-only platform*
- *No offline payments*

2.5 Assumptions and Dependencies

Users have access to modern browsers
Server availability is ensured
Database server remains operational

3. Specific Requirements

This will be the largest and most important section of the SRS. The customer requirements will be embodied within Section 2, but this section will give the D-requirements that are used to guide the project's software design, implementation, and testing.

Each requirement in this section should be:

- *Correct*
- *Traceable (both forward and backward to prior/future artifacts)*
- *Unambiguous*
- *Verifiable (i.e., testable)*
- *Prioritized (with respect to importance and/or stability)*
- *Complete*
- *Consistent*
- *Uniquely identifiable (usually via numbering like 3.4.5.6)*

Attention should be paid to the carefuly organize the requirements presented in this section so that they may easily accessed and understood. Furthermore, this SRS is not the software design document, therefore one should avoid the tendency to over-constrain (and therefore design) the software project within this SRS.

3.1 External Interface Requirements

User Interface – Responsive UI
Hardware – Laptop, Mobile
Software – MERN Stack
Communication – HTTP/HTTPS

3.1.1 User Interfaces

Responsive web UI built with React.js
Dashboard for Admin, Proposer, and User

3.1.2 Hardware Interfaces

Server hosted on cloud/local machine
Client devices: PC, Laptop, Mobile

3.1.3 Software Interfaces

Node.js runtime
MongoDB database
REST APIs

3.1.4 Communications Interfaces

HTTP/HTTPS protocols

3.2 Functional Requirements

- *Secure login system*
- *Product management*
- *Admin approval*
- *Product purchase*

3.2.1 User Registration and Login

Inputs: Name, Email, Password

Processing: Validation and authentication

Outputs: Successful login or error message

Error Handling: Invalid credentials alerts

3.2.2 Product Management

Admin can approve products
Proposers can add/update products
Users can view and purchase products

…

3.5 Non-Functional Requirements

Non-functional requirements may exist for the following attributes. Often these requirements must be achieved at a system-wide level rather than at a unit level. State the requirements in the following sections in measurable terms (e.g., 95% of transaction shall be processed in less than a second, system downtime may not exceed 1 minute per day, > 30 day MTBF value, etc).

3.5.1 Performance

Pages should load within 3 seconds

3.5.2 Reliability

System uptime of 95%

3.5.3 Availability

24/7 availability

3.5.4 Security

Password encryption
Role-based access control

3.5.5 Maintainability

Modular code structure

3.5.6 Portability

Works across major browsers

3.7 Design Constraints

MERN stack usage mandatory
Academic project timeline

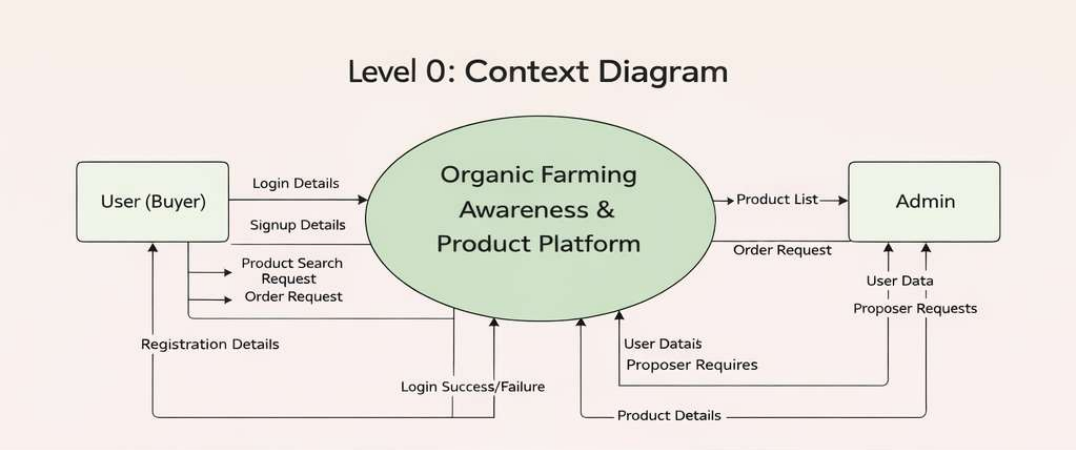
3.9 Other Requirements

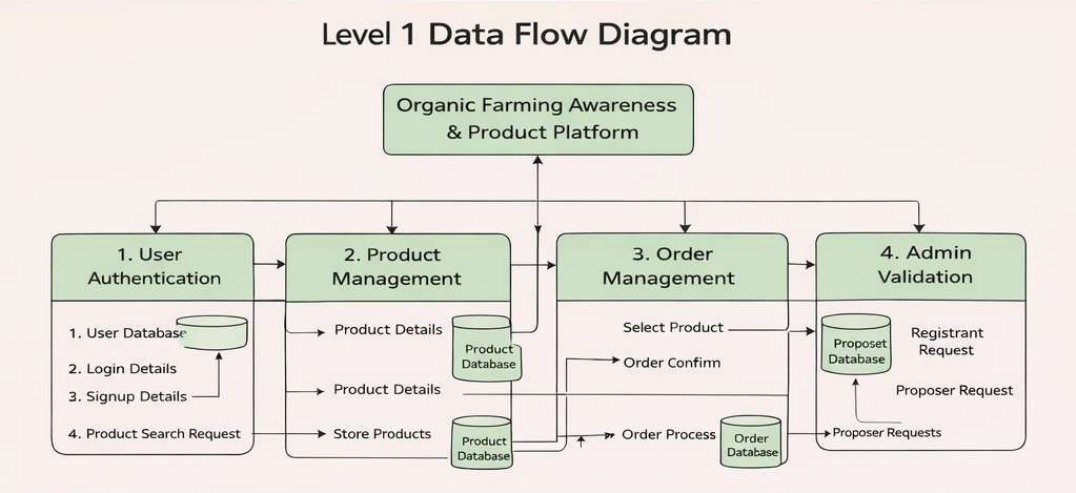
Simple UI for ease of use

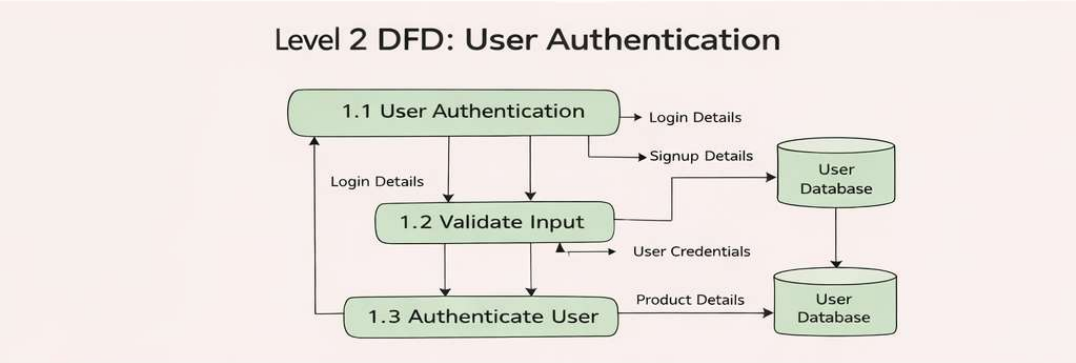
4. Analysis Models

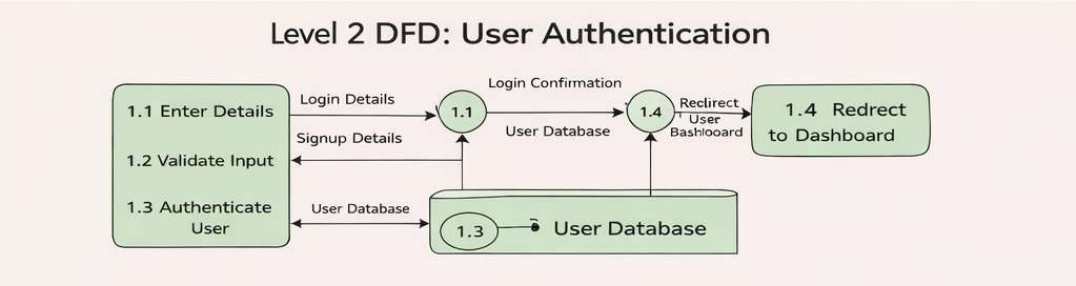
List all analysis models used in developing specific requirements previously given in this SRS. Each model should include an introduction and a narrative description. Furthermore, each model should be traceable the SRS's requirements.

4.1 Data Flow Diagrams (DFD)









A. Appendices

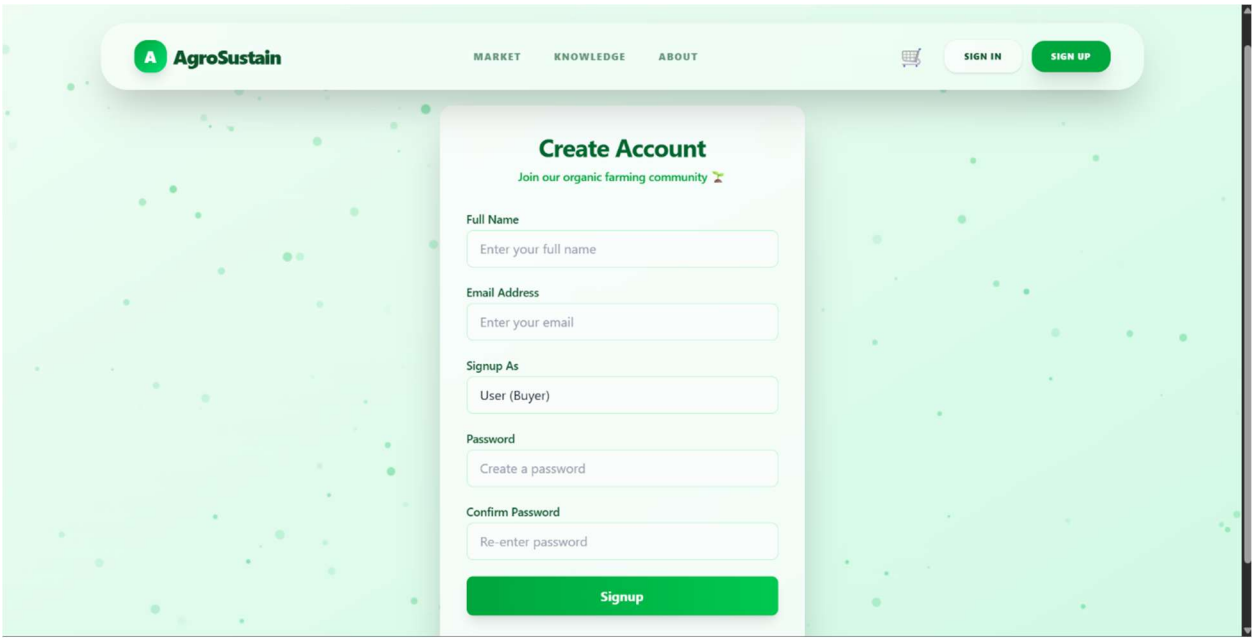
This section contains additional information that supports the Software Requirements Specification (SRS) of the Organic Farming Awareness and Product Platform. The information provided in this section is supplementary and helps in better understanding the system, future scope, and visual representation of the application.

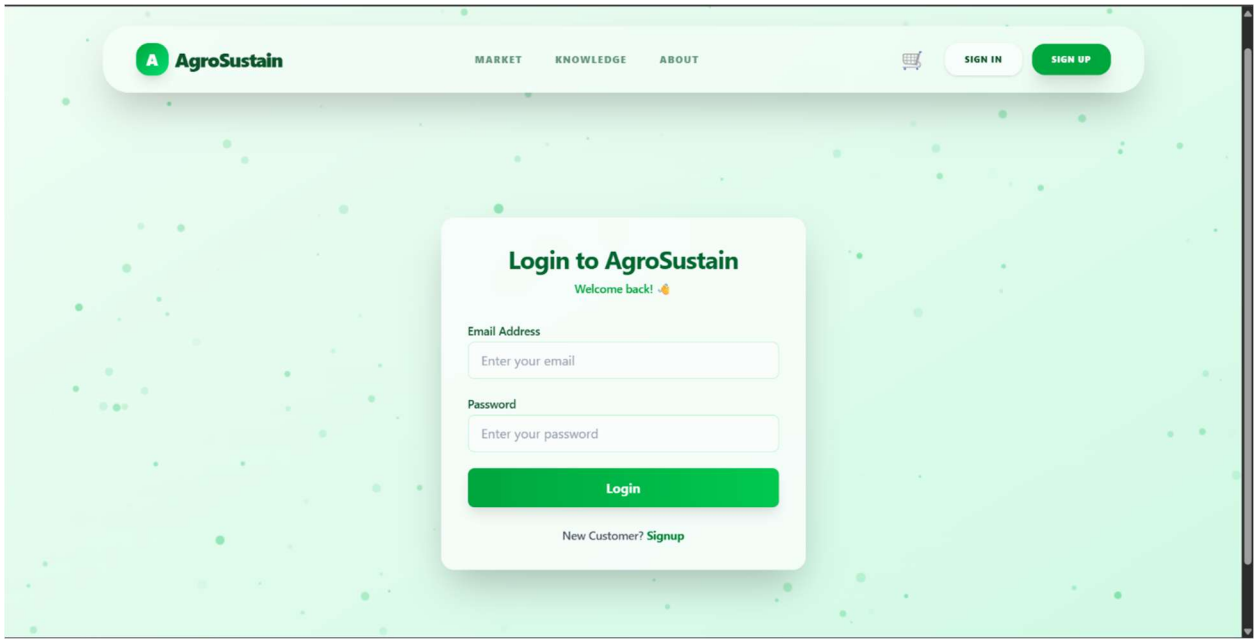
A.1 Future Enhancements

Mobile application
Online payment gateway

A.2 Screenshots

Sign Up Page

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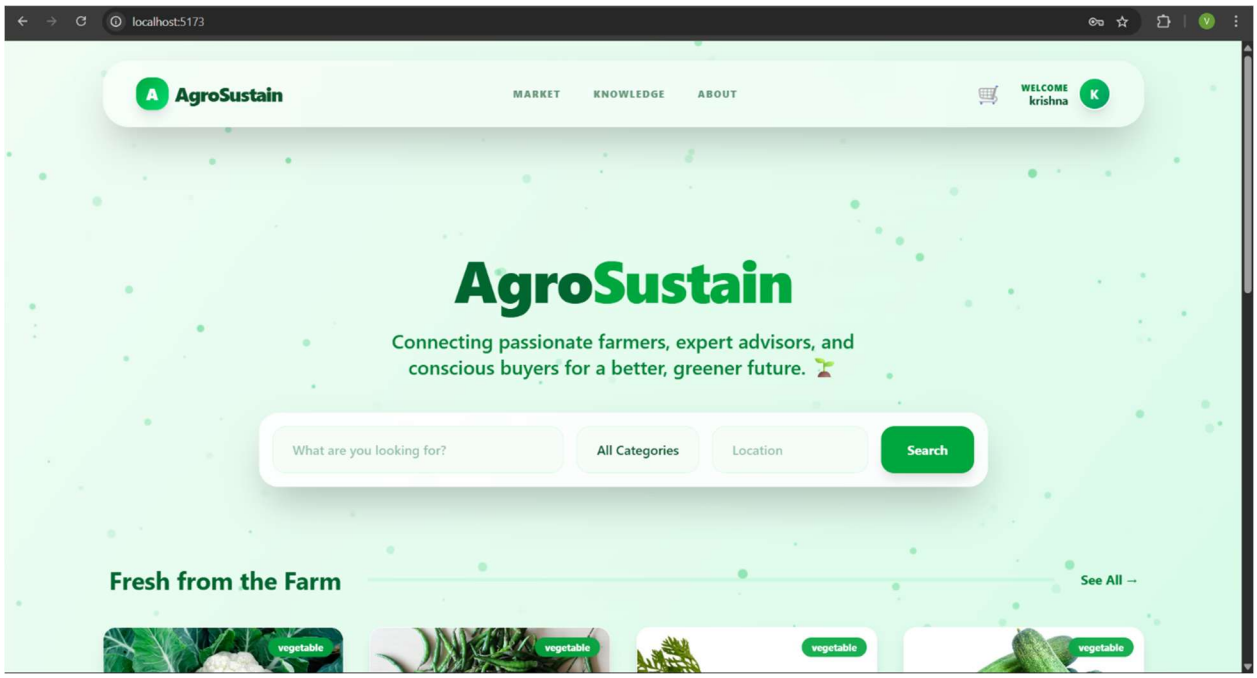
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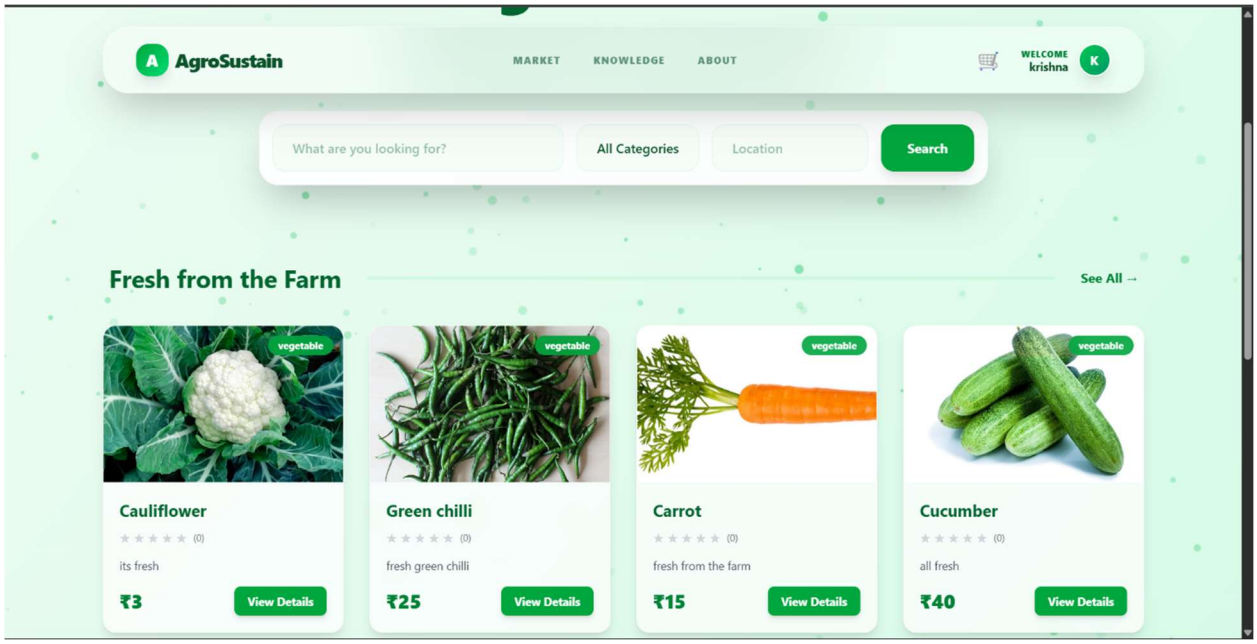
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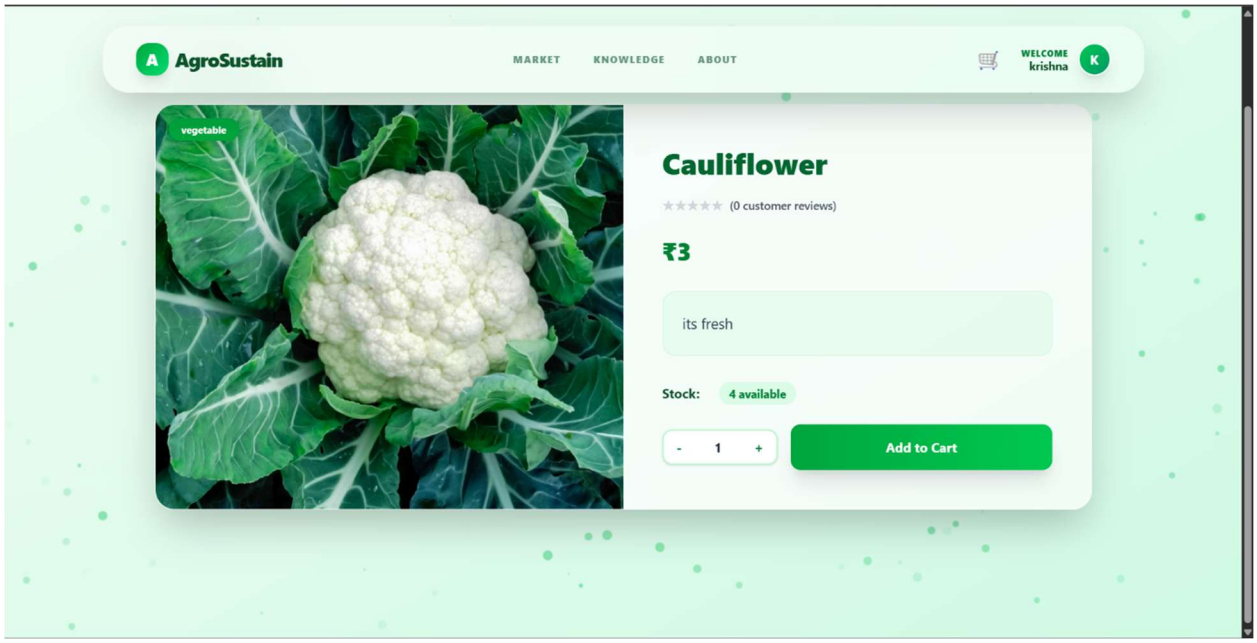
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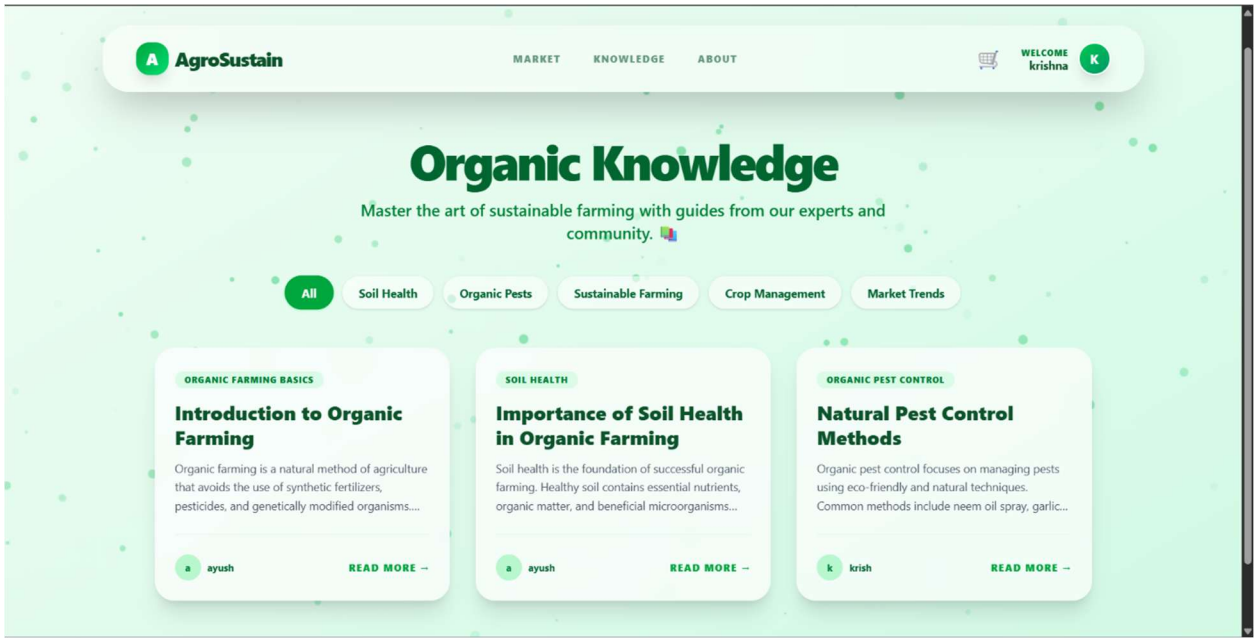
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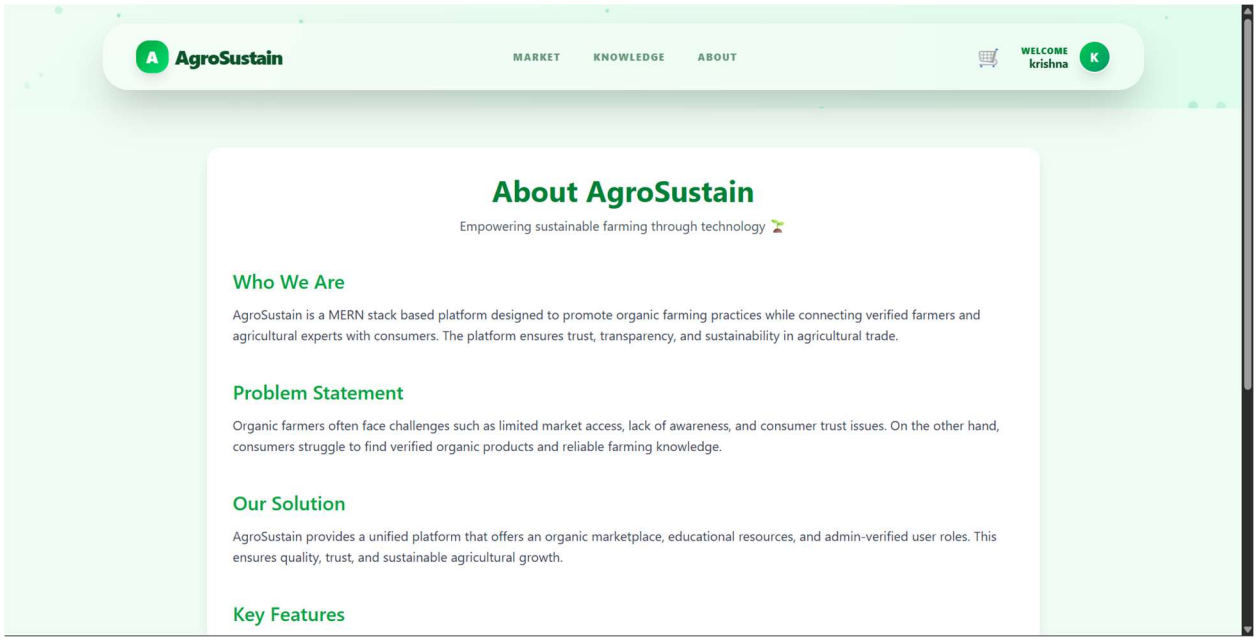
Introduction to Organic Farming
WRITTEN BY
ayush
Organic farming is a natural method of agriculture that avoids the use of synthetic fertilizers, pesticides, and genetically modified organisms. Instead, it relies on natural inputs such as compost, green manure, crop rotation, and biological pest control. The main goal of organic farming is to maintain soil fertility, protect the environment, and produce healthy food.
Organic farming helps improve soil structure, increases biodiversity, and reduces environmental pollution. It is a sustainable approach that benefits farmers,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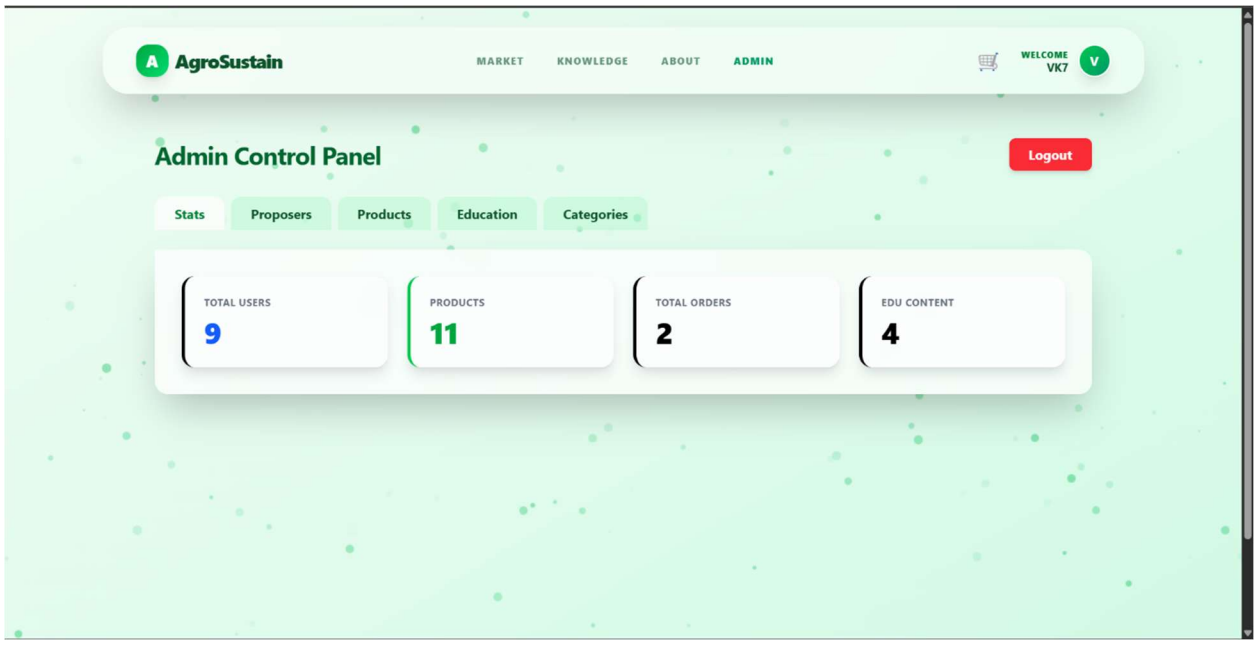
About AgroSustain
Empowering sustainable farming through technology
Who We Are
AgroSustain is a MERN stack based platform designed to promote organic farming practices while connecting verified farmers and agricultural experts with consumers. The platform ensures trust, transparency, and sustainability in agricultural trade.
Problem Statement
Organic farmers often face challenges such as limited market access, lack of awareness, and consumer trust issues. On the other hand, consumers struggle to find verified organic products and reliable farming knowledge.
Our Solution
AgroSustain provides a unified platform that offers an organic marketplace, educational resources, and admin-verified user roles. This ensures quality, trust, and sustainable agricultural growth.
Key Features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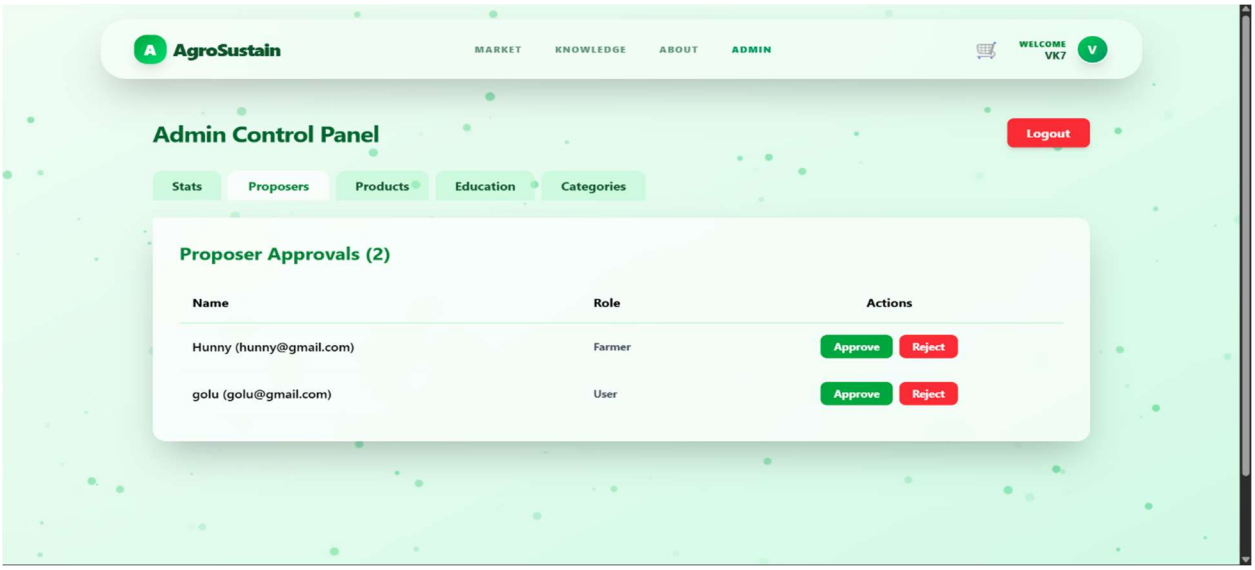
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Farmer
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golu (golu@gmail.com)
User
Approve
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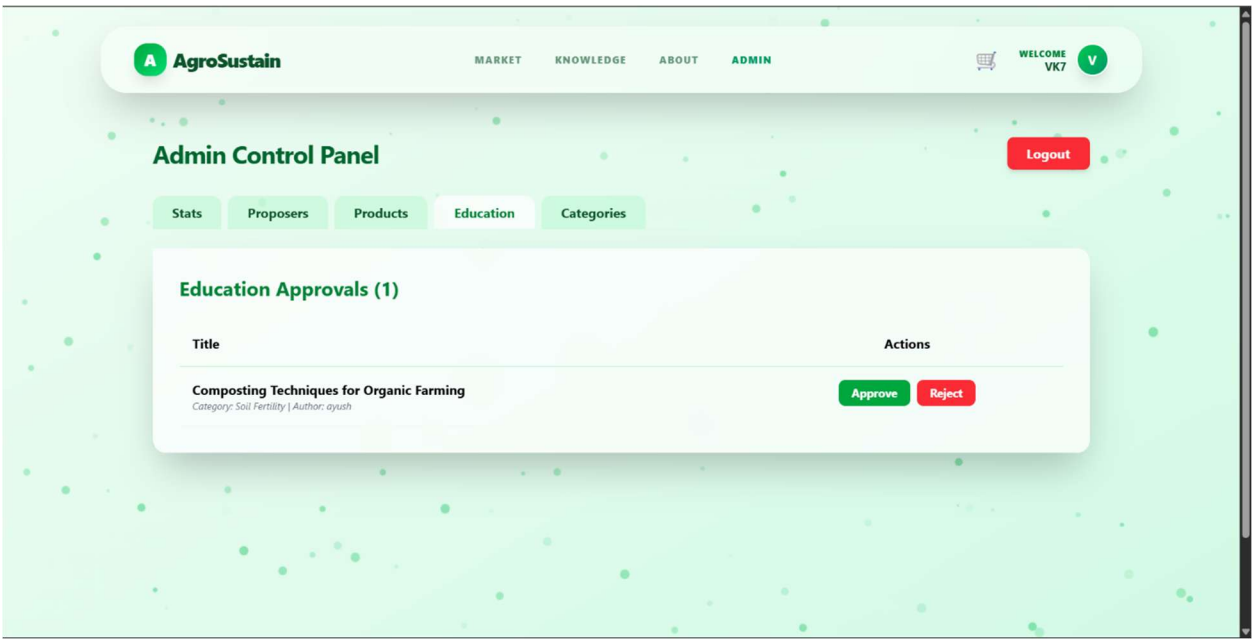
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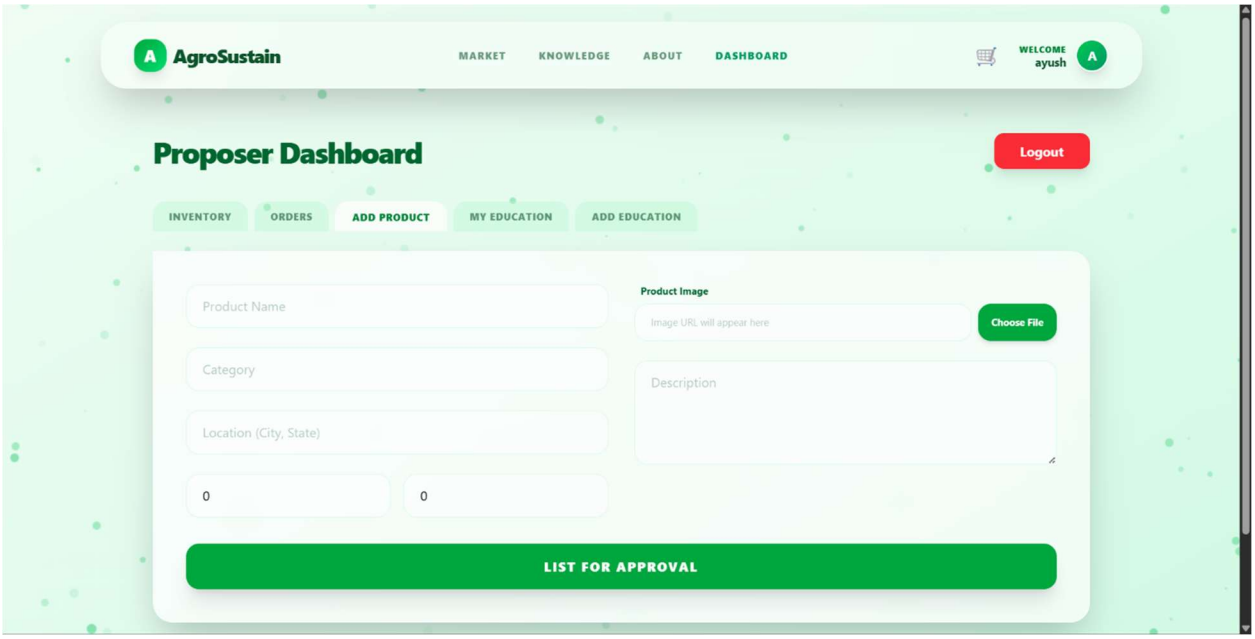

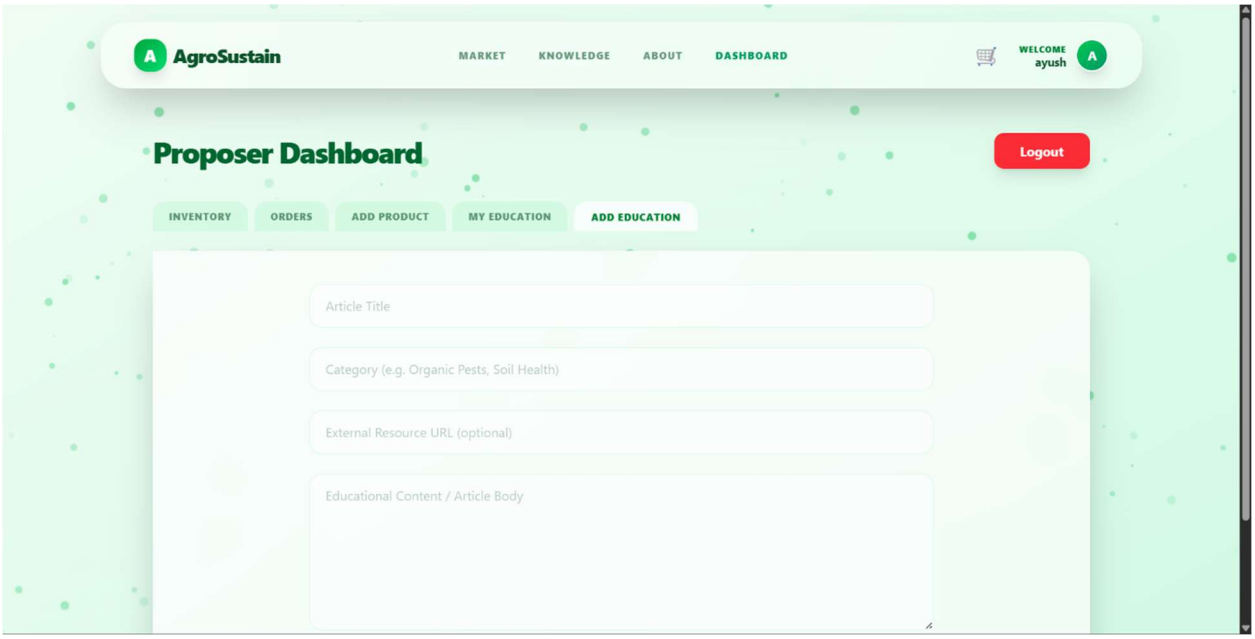

GITHUB: https://github.com/vivek-vk7/AgroSustain

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